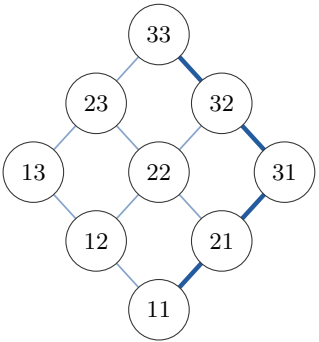
(a) Synchronized lift

$u_{ij} = (\alpha_i + \beta_j)/2$



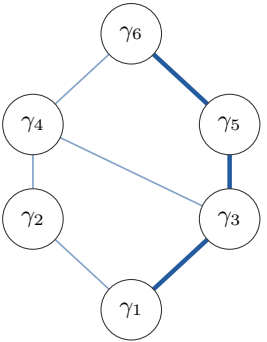
Six saturated bottom-to-top paths

All six means are $11/20 = 0.55$

Highlighted: one of six exact optimizers

(b) Chain–diamond pullback

Asynchronous advancement



Three source-to-target paths

Unique optimum: $\gamma_1 \prec \gamma_3 \prec \gamma_5 \prec \gamma_6$

Exact mean: 0.7078125

$(\alpha_1, \alpha_2, \alpha_3) = (.20, .50, .80), \quad (\beta_1, \beta_2, \beta_3) = (.30, .60, .90)$

Both examples use zero transition costs; full utility vectors and path enumerations are provided in Online Resource 1